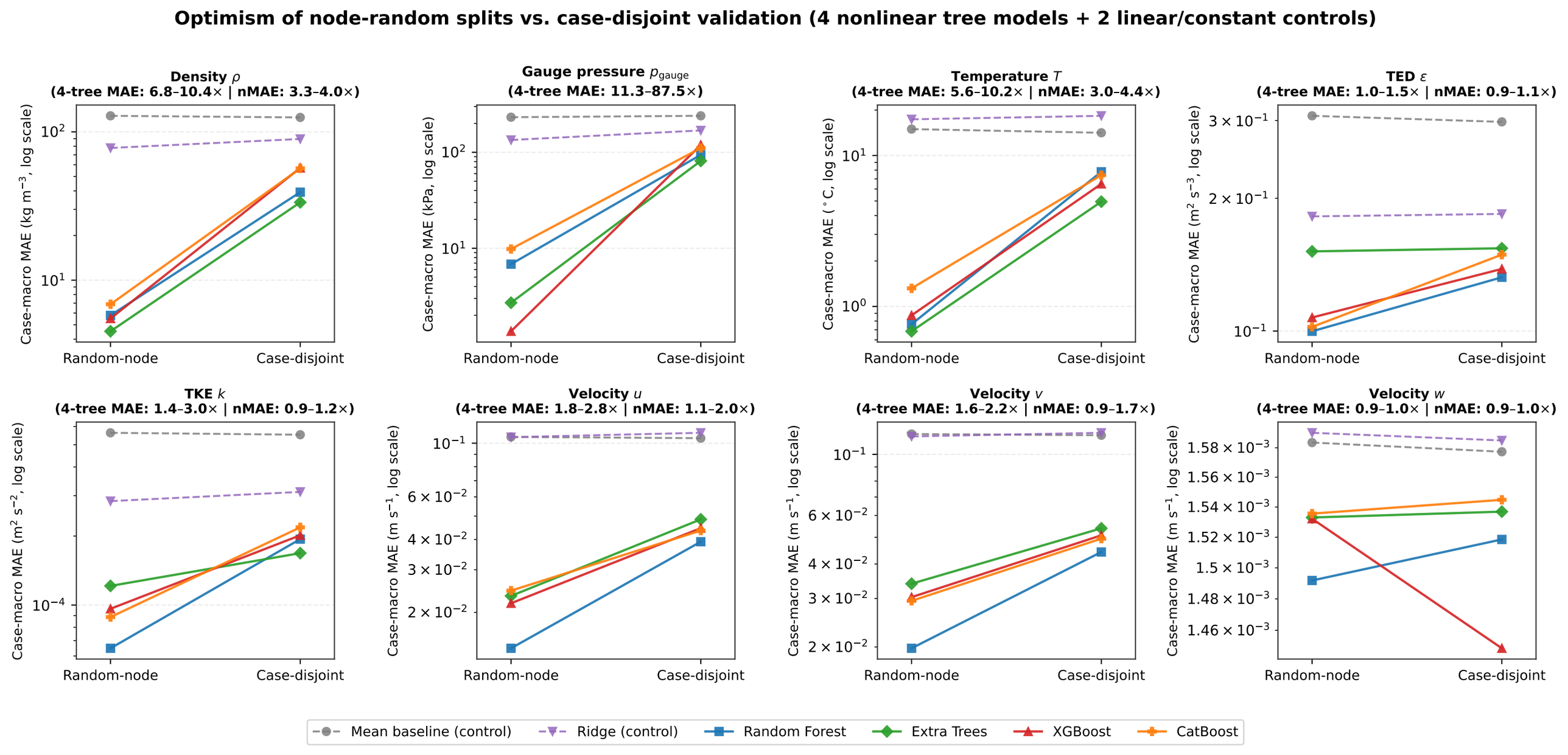
Optimism of node-random splits vs. case-disjoint validation (4 nonlinear tree models + 2 linear/constant controls)
Density ρ
(4-tree MAE: 6.8–10.4× | nMAE: 3.3–4.0×)
Case-macro MAE (kg m⁻³, log scale)
Gauge pressure p_gauge
(4-tree MAE: 11.3–87.5×)
Case-macro MAE (kPa, log scale)
Temperature T
(4-tree MAE: 5.6–10.2× | nMAE: 3.0–4.4×)
Case-macro MAE (°C, log scale)
TED ε
(4-tree MAE: 1.0–1.5× | nMAE: 0.9–1.1×)
Case-macro MAE (m² s⁻³, log scale)
TKE k
(4-tree MAE: 1.4–3.0× | nMAE: 0.9–1.2×)
Case-macro MAE (m² s⁻², log scale)
Velocity u
(4-tree MAE: 1.8–2.8× | nMAE: 1.1–2.0×)
Case-macro MAE (m s⁻¹, log scale)
Velocity v
(4-tree MAE: 1.6–2.2× | nMAE: 0.9–1.7×)
Case-macro MAE (m s⁻¹, log scale)
Velocity w
(4-tree MAE: 0.9–1.0× | nMAE: 0.9–1.0×)
Case-macro MAE (m s⁻¹, log scale)
Random-node
Case-disjoint
Mean baseline (control)
Ridge (control)
Random Forest
Extra Trees
XGBoost
CatBoost